



HOW TO CONFIGURE MPLS VPN

STEP-BY-STEP CONFIGURATION GUIDE



PRACTICAL
EXAMPLES



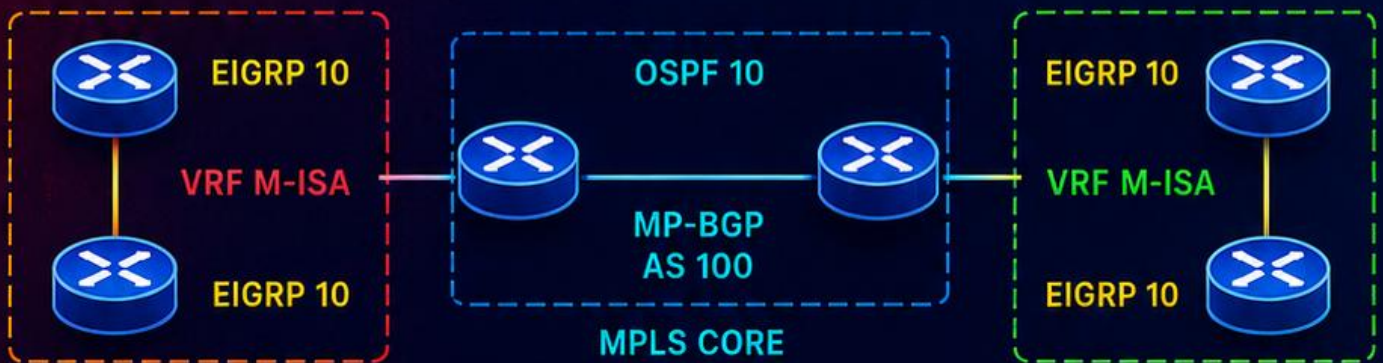
NETWORK
TOPOLOGY



SECURE
CONNECTIVITY



VERIFICATION
STEPS



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NETWORKING



SECURE



SCALABLE



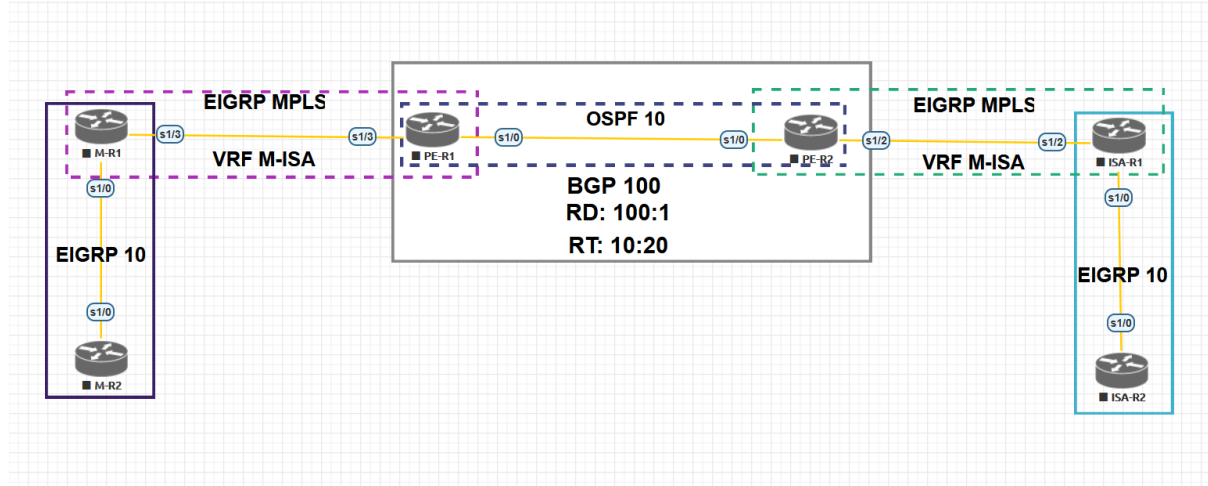
RELIABLE



CONNECTED

How to Configure MPLS VPN

Network Topology:



Address Table:

Device	Interface	IP Address	Subnet Mask
M-R1	S1/0	172.22.1.2	255.255.255.252
	S1/3	172.22.1.34	255.255.255.252
	Loopback 1	1.1.1.1	255.255.255.255
M-R2	S1/0	172.22.1.1	255.255.255.252
	Loopback 1	10.10.10.10	255.255.255.255
PE-R1	S1/0	172.20.0.1	255.255.255.252
	S1/3	172.22.1.33	255.255.255.252
	Loopback 0	172.16.10.10	255.255.255.255
PE-R2	S1/0	172.20.0.2	255.255.255.252
	S1/2	172.22.1.17	255.255.255.252
	Loopback 0	172.16.10.11	255.255.255.255
ISA-R1	S1/0	172.22.1.25	255.255.255.255
	S1/2	172.22.1.18	255.255.255.252
	Loopback 1	2.2.2.2	255.255.255.255
ISA-R2	S1/0	172.22.1.26	255.255.255.252
	Loopback 1	20.20.20.20	255.255.255.255

Step 1: Configure IGB between the M-R1 & M-R2:

```
M-R1(config)#Router eigrp 10  
M-R1(config-router)# network 1.1.1.1 0.0.0.0  
M-R1(config-router)# network 172.22.1.0 0.0.0.3  
M-R1(config-router)# network 172.22.1.32 0.0.0.3
```

```
M-R2(config)#Router eigrp 10  
M-R2(config-router)#network 10.10.10.10 0.0.0.0  
M-R2(config-router)#network 172.22.1.0 0.0.0.3
```

Step 2: Configure IGB between the ISA-R1 & ISA-R2:

```
ISA-R1(config)#Router eigrp 10  
ISA-R1(config-router)# network 2.2.2.2 0.0.0.0  
ISA-R1(config-router)# network 172.22.1.16 0.0.0.3  
ISA-R1(config-router)# network 172.22.1.24 0.0.0.3
```

```
ISA-R2(config)#Router eigrp 10  
ISA-R2(config-router)# network 20.20.20.20 0.0.0.0  
ISA-R2(config-router)# network 172.22.1.24 0.0.0.3
```

Step 3: Configure IGP in the PE Routers

```
PE-R1(config)#router ospf 10  
PE-R1(config-router)# network 172.16.10.10 0.0.0.0 area 0
```

```
PE-R1(config-router)# network 172.20.0.0 0.0.0.3 area 0
```

Do the same for PE-R2

```
PE-R2(config)#router ospf 10
```

```
PE-R2(config-router)# network 172.16.10.11 0.0.0.0 area 0
```

```
PE-R2(config-router)# network 172.20.0.0 0.0.0.3 area 0
```

Step 4: Configure MP-BGP Between PE Routers:

```
PE-R1(config)#router bgp 100
```

```
PE-R1(config-router)# no bgp default ipv4-unicast
```

```
PE-R1(config-router)# neighbor 172.16.10.11 remote-as 100
```

```
PE-R1(config-router)# neighbor 172.16.10.11 update-source Loopback0
```

```
PE-R1(config-router)# address-family ipv4 unicast
```

```
PE-R1(config-router-af)# neighbor 172.16.10.11 activate
```

```
PE-R2(config)#router bgp 100
```

```
PE-R2(config-router)# no bgp default ipv4-unicast
```

```
PE-R2(config-router)# neighbor 172.16.10.10 remote-as 100
```

```
PE-R2(config-router)# neighbor 172.16.10.10 update-source Loopback0
```

```
PE-R2(config-router)#address-family ipv4 unicast
```

```
PE-R2(config-router-af)# neighbor 172.16.10.10 activate
```


Step 5: Configure MPLS on PE Routers

```
PE-R1(config)# mpls ldp router-id loopback0 force
```

This command will **force** the MPLS to use the **loopback 0**

```
PE-R1(config)# interface s1/0
```

```
PE-R1(config-if)# mpls ip
```

This command will **enable** the MPLS inside the interface

```
PE-R1(config)# ip vrf M-ISA
```

This command will create the VRF that will separate each customer

```
PE-R1(config-vrf)# rd 100:1
```

```
PE-R1(config-vrf)# route-target 10:20
```

The rd, think about it like a passport, it has to be unique between the customers and the Route target, like the visa is, it has to be the same between the customers. rd target has to have the same ID, both ends, otherwise it won't work

```
PE-R1(config)# interface s1/3
```

```
PE-R1(config-if)# ip vrf forwarding M-ISA
```

```
PE-R1(config-if)# ip add 172.22.1.33 255.255.255.252
```

This command will enable the vrf into the interface, but when you do it, the vrf will clear your interface IP, so you have to re-enter it again.

Do the same for PE-R2

```
PE-R2(config)# mpls ldp router-id loopback0 force
```

```
PE-R2(config)# interface s1/0
```

```
PE-R2(config-if)# mpls ip
```

```
PE-R2(config)# ip vrf M-ISA
```

```
PE-R2(config-vrf)# rd 100:1
```

```
PE-R2(config-vrf)# route-target 10:20
```

```
PE-R2(config)# interface s1/2
```

```
PE-R2(config-if)# ip vrf forwarding M-ISA
```

```
PE-R2(config-if)# ip add 172.22.1.17 255.255.255.252
```

Step 6: Configure EIGRP in the PE routers for the VRF to communicate with the customer routers.

```
PE-R1(config)# router eigrp MPLS
```

```
PE-R1(config-router)# address-family ipv4 vrf M-ISA autonomous-system 10
```

```
PE-R1(config-router-af)# network 172.22.1.32 255.255.255.252
```

This command **will establish communication** with the customer routers

Do the same for PE-R2

```
PE-R2(config)# router eigrp MPLS
```

```
PE-R2(config-router)# address-family ipv4 vrf M-ISA autonomous-system 10
```

```
PE-R2(config-router-af)# network 172.22.1.16 255.255.255.252
```

Step 7: Configure the VPNv4 in the BGP to send and receive from the VRF

```
PE-R1(config)#router bgp 100
```

```
PE-R1(config-router)# address-family vpnv4
```

```
PE-R1(config-router-af)# neighbor 172.16.10.11 activate
```

```
PE-R1(config-router-af)# neighbor 172.16.10.11 send-community both
```

Inside the VPNv4 you will make the bgp import and export the vrf from the customer.

```
PE-R1(config-router-af)# address-family ipv4 vrf M-ISA
```

```
PE-R1(config-router-af)# redistribute eigrp 10
```

Now we will redistribute EIGRP inside the BGP to share the networks that have been learned from the VRF

Do the same for PE-R2

```
PE-R2(config)# router bgp 100
```

```
PE-R2(config-router)# address-family vpnv4
```

```
PE-R2(config-router-af)# neighbor 172.16.10.10 activate
```

```
PE-R2(config-router-af)# neighbor 172.16.10.10 send-community both
```

```
PE-R2(config-router-af)# address-family ipv4 vrf M-ISA
```

```
PE-R2(config-router-af)# redistribute eigrp 10
```

Step 8: Go to the EIGRP and redistribute the BGP table in it

```
PE-R1(config)# router eigrp MPLS
```

```
PE-R1(config-router)# address-family ipv4 vrf M-ISA autonomous-system 10
```

```
PE-R1(config-router-af)# topology base
```

```
PE-R1(config-router-af-topology)# redistribute bgp 100 metric 64 1000 255 1 1500
```

Do not **forget** the metric for the EIGRP because it will set the network infinity.

Do the same for PE-R2

```
PE-R2(config)# router eigrp MPLS
```

```
PE-R2(config-router)# address-family ipv4 vrf M-ISA autonomous-system 10
```

```
PE-R2(config-router-af)# topology base
```

PE-R2(config-router-af-topology)# redistribute bgp 100 metric 64 1000 255 1 1500

Step 9: Verify Configuration

To check if you did everything correctly, go to the customer router and enter **show ip route**, and you should have the IP addresses from the other end.

```
M-R2
M-R1>en
M-R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override

Gateway of last resort is not set

 1.0.0.0/32 is subnetted, 1 subnets
D    1.1.1.1 [90/2297856] via 172.22.1.2, 01:09:06, Serial1/0
D    2.0.0.0/32 is subnetted, 1 subnets
D    2.2.2.2 [90/3833856] via 172.22.1.2, 01:07:56, Serial1/0
 10.0.0.0/32 is subnetted, 1 subnets
C    10.10.10.10 is directly connected, Loopback1
D    20.0.0.0/32 is subnetted, 1 subnets
D    20.20.20.20 [90/3321856] via 172.22.1.2, 01:07:56, Serial1/0
 172.22.0.0/16 is variably subnetted, 5 subnets, 2 masks
C    172.22.1.0/30 is directly connected, Serial1/0
L    172.22.1.1/32 is directly connected, Serial1/0
D    172.22.1.16/30 [90/3193856] via 172.22.1.2, 01:07:56, Serial1/0
D    172.22.1.24/30 [90/3705856] via 172.22.1.2, 01:07:56, Serial1/0
D    172.22.1.32/30 [90/2681856] via 172.22.1.2, 01:09:04, Serial1/0
M-R1#
```

```
M-R2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override

Gateway of last resort is not set

 1.0.0.0/32 is subnetted, 1 subnets
D    1.1.1.1 [90/2297856] via 172.22.1.2, 01:09:06, Serial1/0
D    2.0.0.0/32 is subnetted, 1 subnets
D    2.2.2.2 [90/3833856] via 172.22.1.2, 01:07:56, Serial1/0
 10.0.0.0/32 is subnetted, 1 subnets
C    10.10.10.10 is directly connected, Loopback1
D    20.0.0.0/32 is subnetted, 1 subnets
D    20.20.20.20 [90/3321856] via 172.22.1.2, 01:07:56, Serial1/0
 172.22.0.0/16 is variably subnetted, 5 subnets, 2 masks
C    172.22.1.0/30 is directly connected, Serial1/0
L    172.22.1.1/32 is directly connected, Serial1/0
D    172.22.1.16/30 [90/3193856] via 172.22.1.2, 01:07:56, Serial1/0
D    172.22.1.24/30 [90/3705856] via 172.22.1.2, 01:07:56, Serial1/0
D    172.22.1.32/30 [90/2681856] via 172.22.1.2, 01:09:04, Serial1/0
M-R1#traceroute 20.20.20.20
Type escape sequence to abort.
Tracing the route to 20.20.20.20
VRF info: (vrf in name/id, vrf out name/id)
 1 172.22.1.2 10 msec 10 msec 10 msec
 2 172.22.1.33 21 msec 20 msec 18 msec
 3 172.22.1.17 [MPLS: Label 18 Exp 0] 30 msec 28 msec 30 msec
 4 172.22.1.18 40 msec * 39 msec
M-R1#
```